

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
Second Semester B.Tech Degree Examination June 2022 (2019 scheme)

Course Code: EST110
Course Name: ENGINEERING GRAPHICS
(2019 -Scheme)

Max. Marks: 100

Duration: 3 Hours

Instructions: Retain Construction lines. Show necessary dimensions. Answer any ONE question from each module. Each question carries 20 marks.

MODULE 1

- 1 One end point of a line AB is 12 mm above HP and is 15 mm in-front of VP. Other end point is 50 mm above HP and is 42 mm in front of VP. Draw the projections of the line AB if its elevation measures 70 mm. Find out its true length and the true inclinations with respect to the reference planes.
- 2 One end point P of a line PQ, 75 mm long, is 10 mm above HP and 20 mm in front of VP. The line is inclined 45° to HP and its plan is inclined 35° to x-y line. Draw the projections of the line PQ and find out true inclination of the line with respect to VP.

MODULE 2

- 3 A hexagonal prism base 20 mm side and axis 40 mm long is placed with one of its base edges on the HP such that the axis is inclined at 30° to HP and 45° to VP. Draw the projections of the prism.
- 4 A cone of base diameter 50 mm and axis length 60 mm is resting on VP on one of its generators with the front view of the axis inclined at 40° to HP. Draw its projections.

MODULE 3

- 5 A hexagonal pyramid, side of base 25 mm and altitude 70 mm long, rests with its base on HP with two of its base sides parallel to VP. It is cut by a section plane perpendicular to VP, inclined at 45° to HP and passing through the axis 15mm from the base. Draw the sectional top view and true shape of the section.
- 6 A pentagonal pyramid, side of base 50 mm and height 80 mm rests on its base on the ground with one of its base sides parallel to VP. A section plane perpendicular to VP

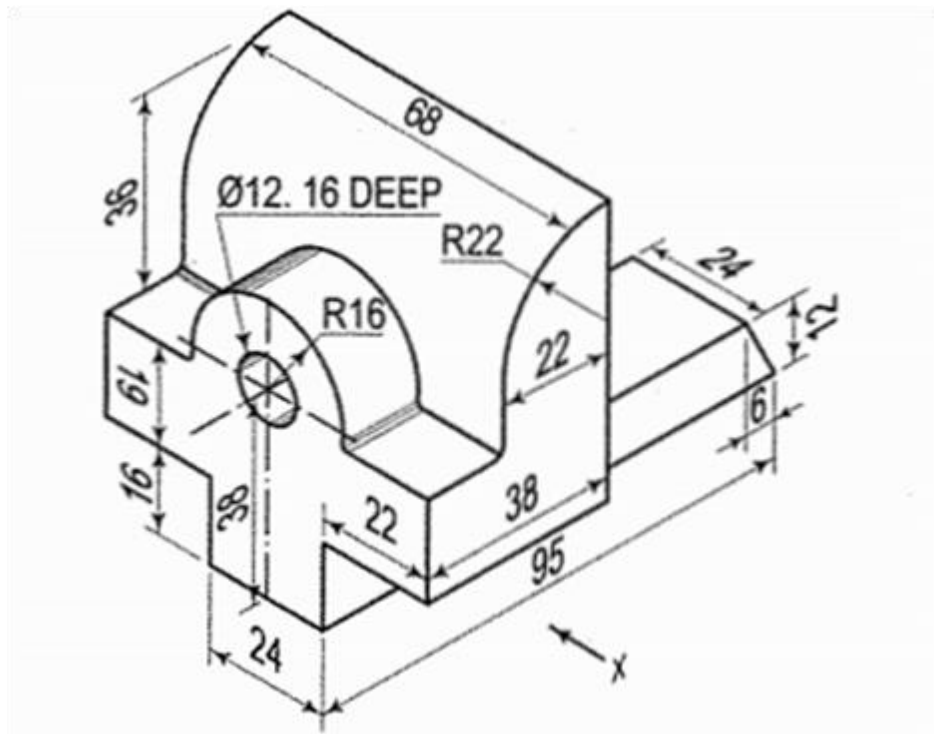
and inclined at 30° to HP cuts the pyramid, bisecting its axis. Draw the development of the truncated pyramid.

MODULE 4

- 7 A square pyramid of base edge 20 mm and height 40 mm is mounted centrally on a face of a cube of base edge 50 mm. Draw the isometric projection of the objects.
- 8 Draw the isometric projections of a hexagonal prism with edge of base 30 mm and axis 60 mm when it rests on a rectangular face on the ground.

MODULE 5

- 9 A square pyramid of base side 30 mm and height 45 mm is resting on the ground plane. The nearest edge of the base is parallel to and 20 mm behind the Picture Plane (PP). The station point is situated at a distance of 70 mm in front of the PP, 40 mm to the right of the axis of the pyramid, and 60 mm above the ground. Draw the perspective view of the pyramid.
- 10 Draw the front view, top view, and side view of the object given below. Front view should be drawn as seen in the direction of the arrow X.

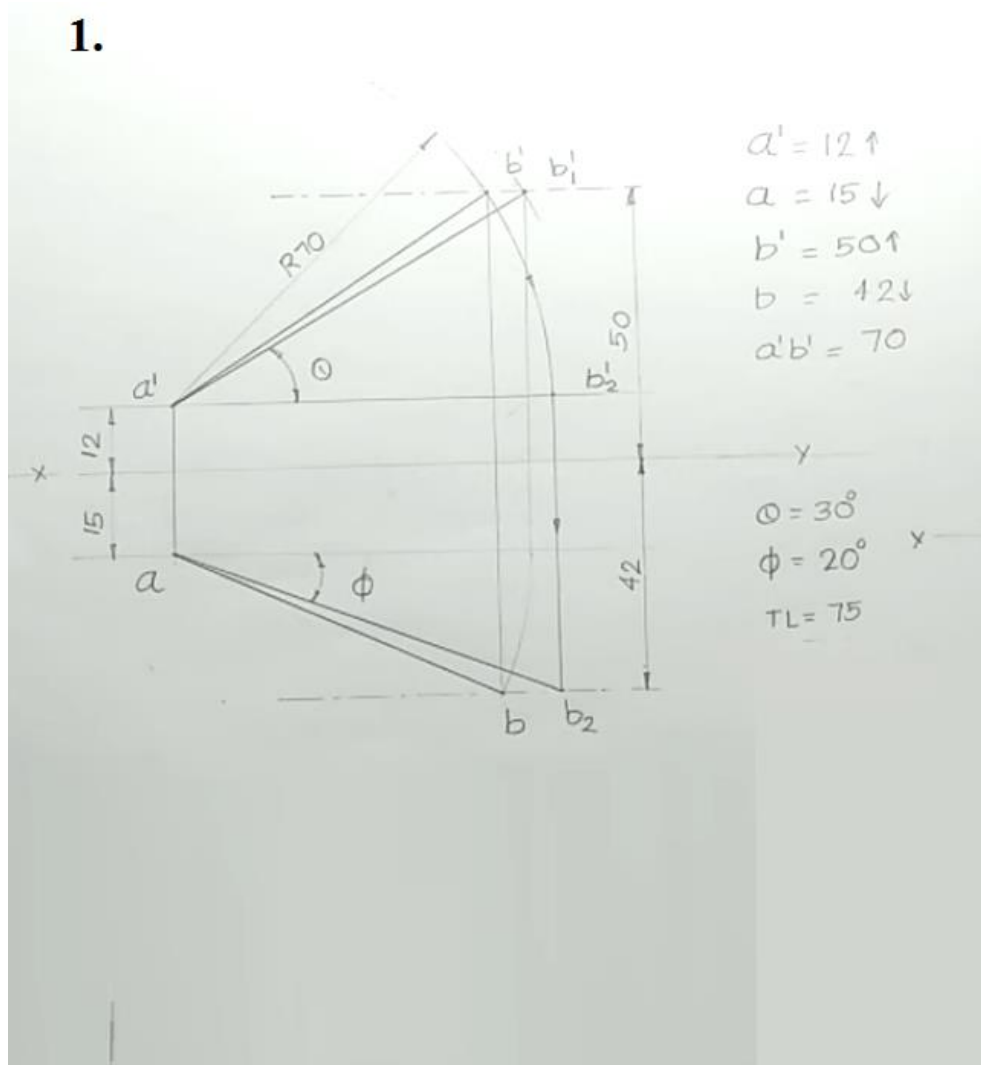


(5 x 20 = 100 Marks)

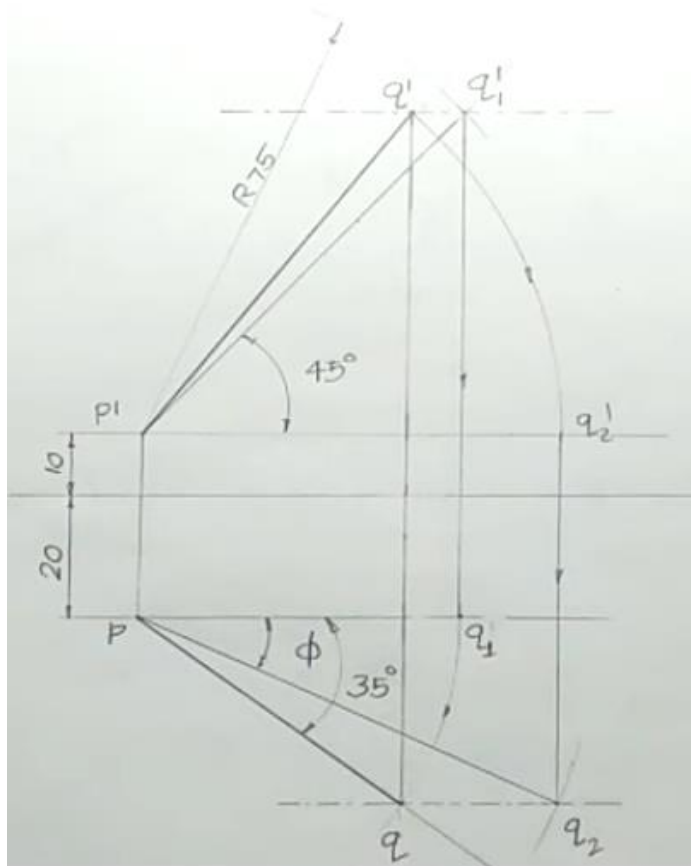
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
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ANSWER KEY

1.



2.



$$PQ, TL, = 75 = P'q_1' = Pq_2$$

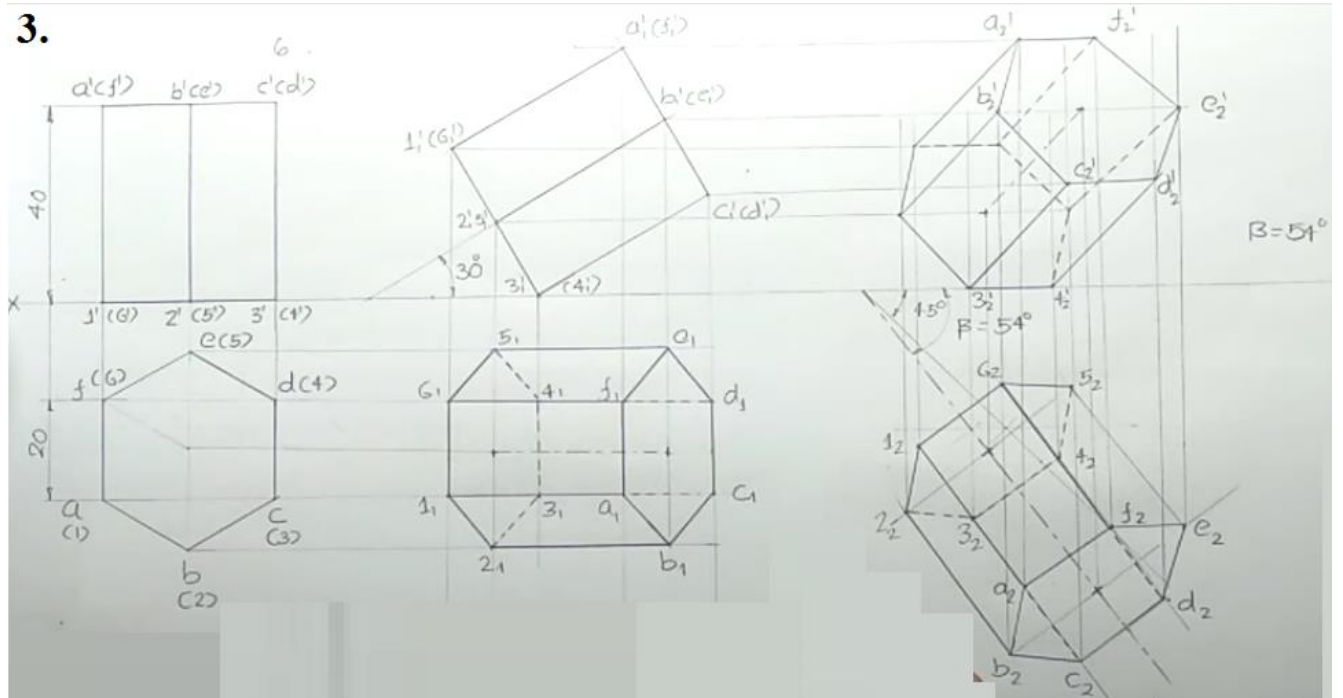
$$P' = 101$$

$$P = 201$$

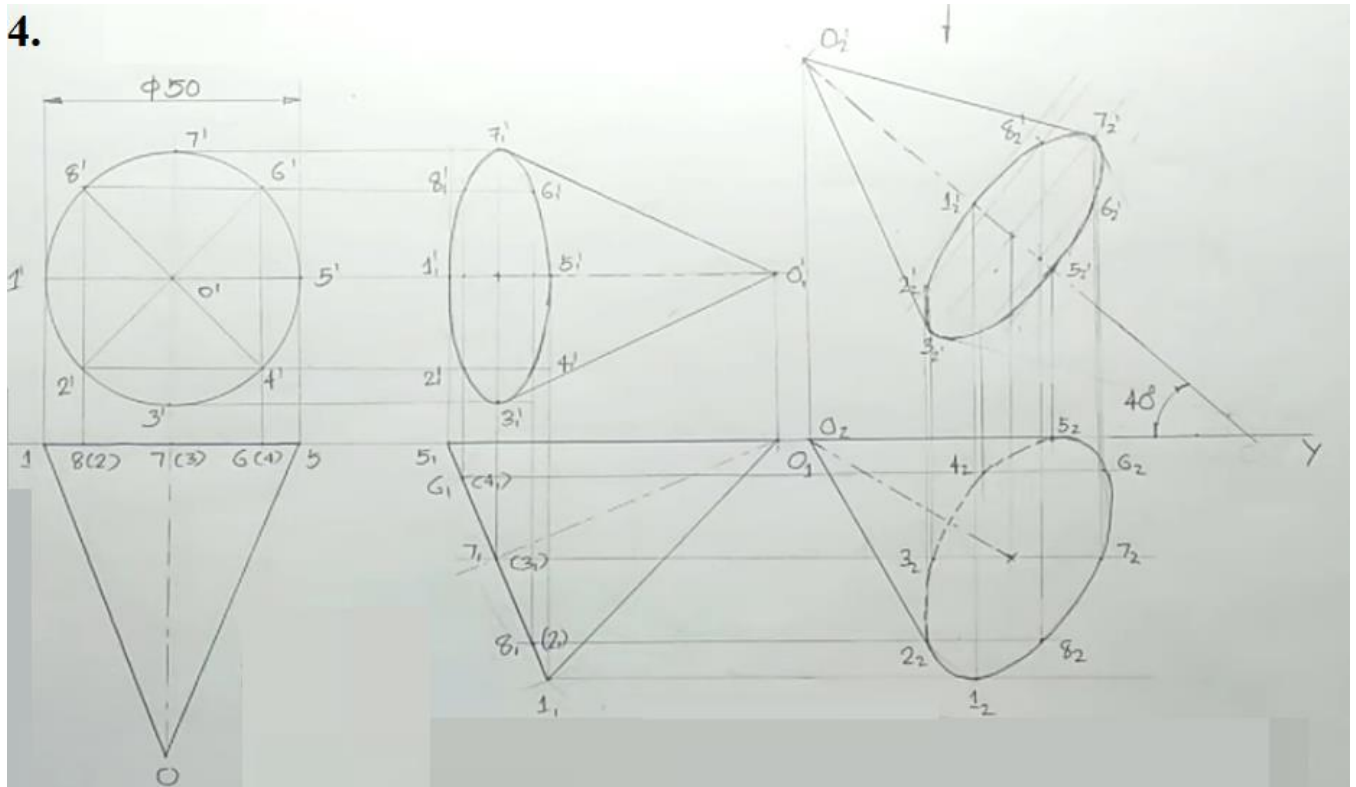
$$\alpha = 45^\circ$$

$$\beta = 35^\circ$$

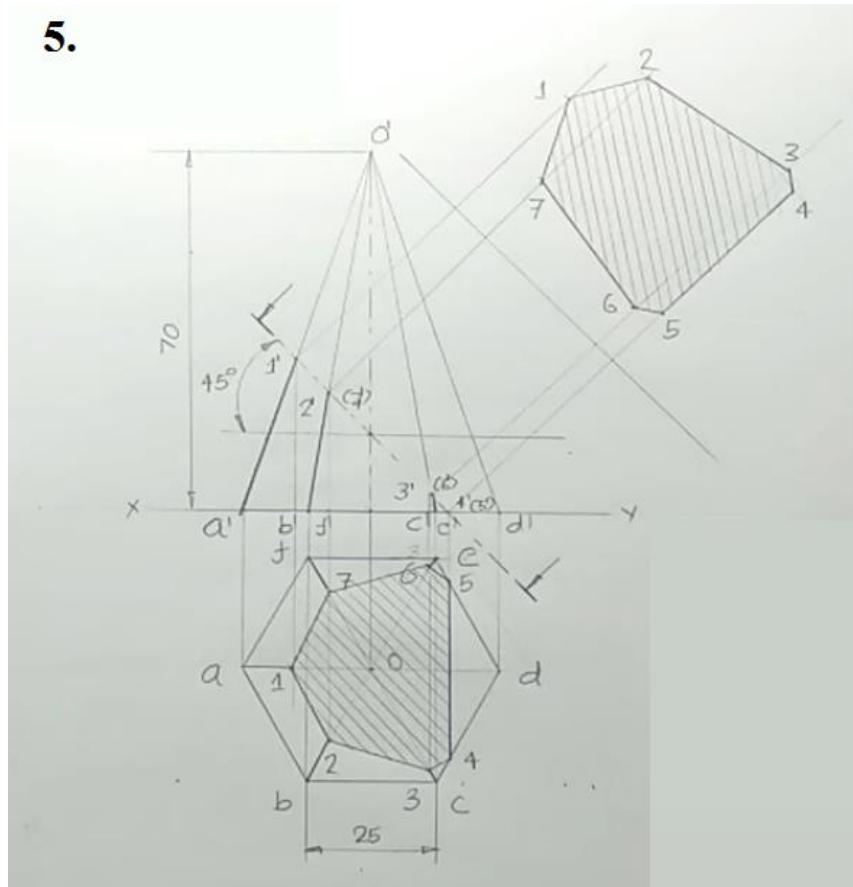
3.



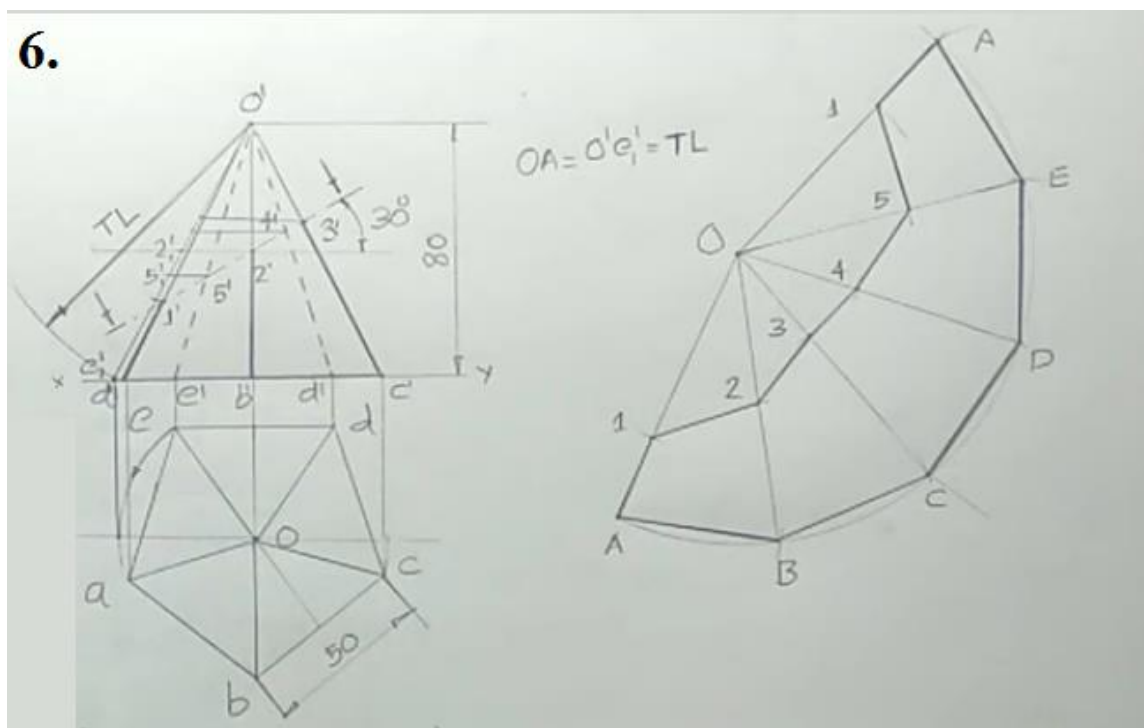
4.



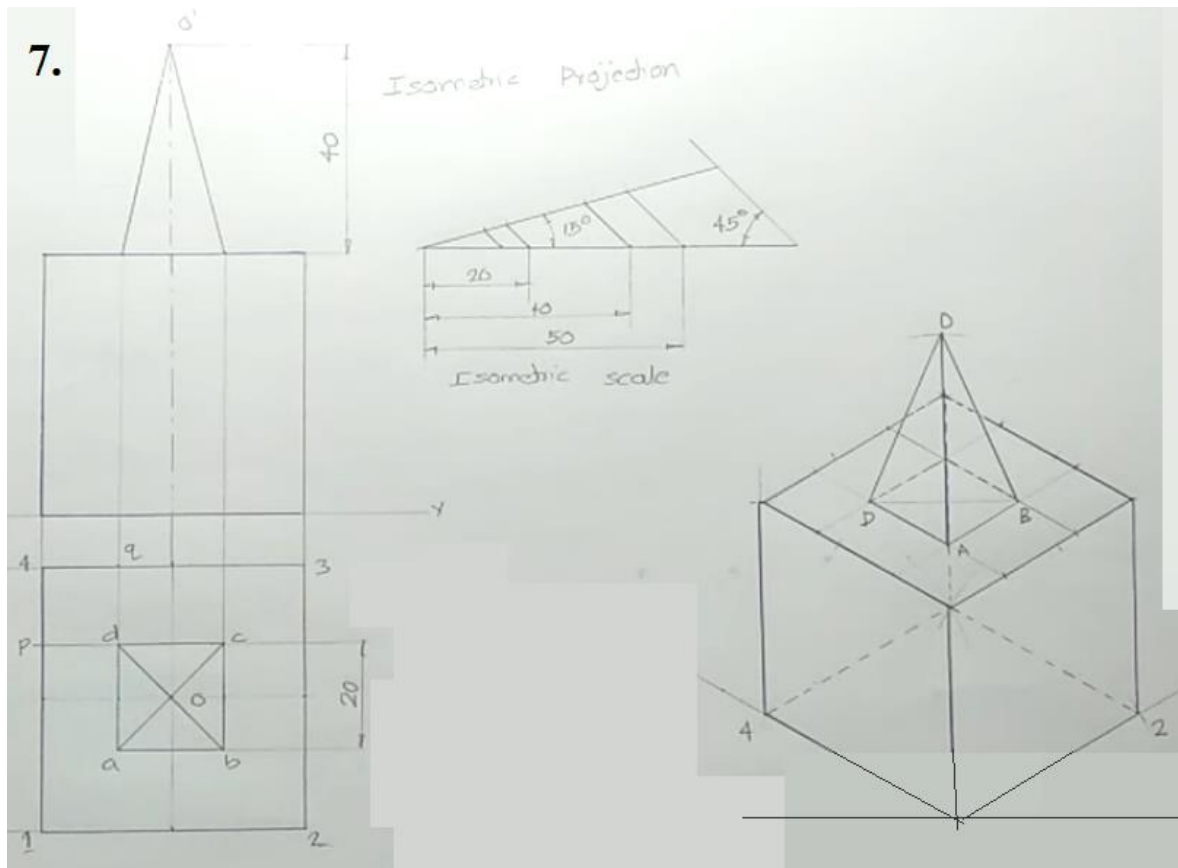
5.



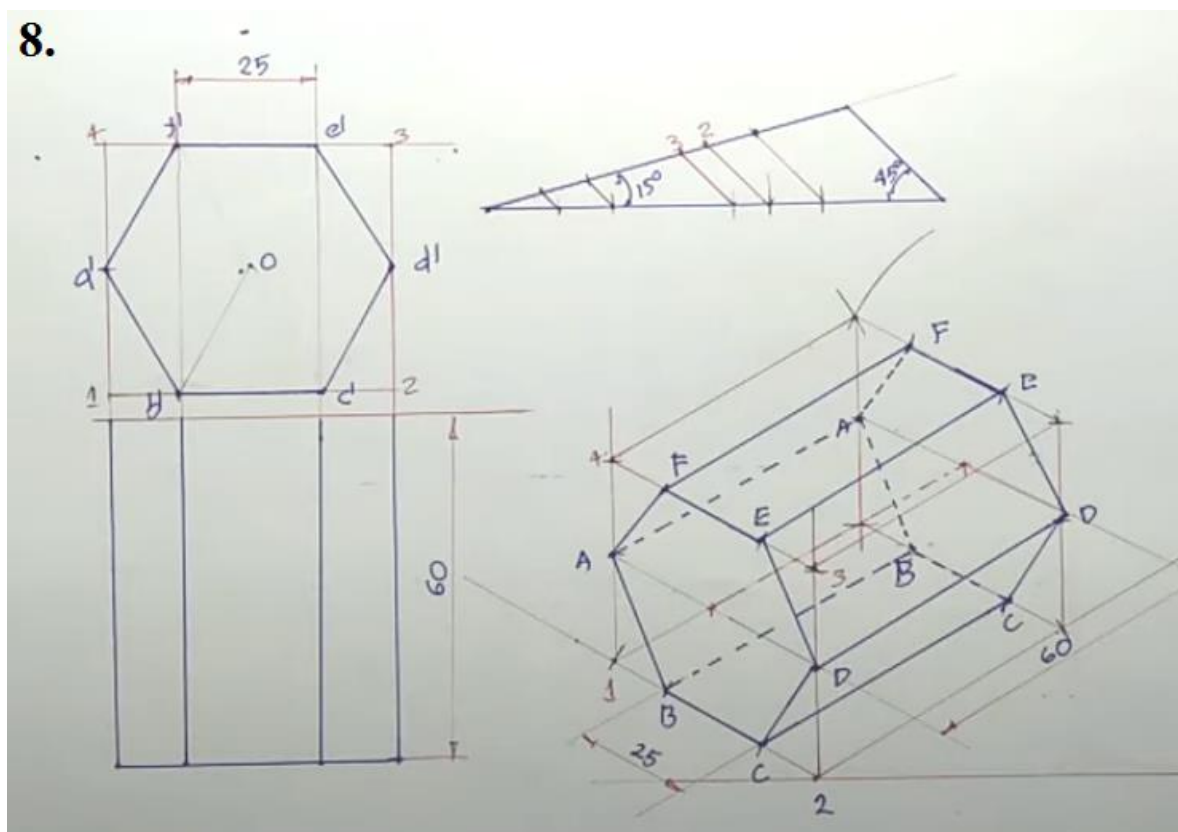
6.



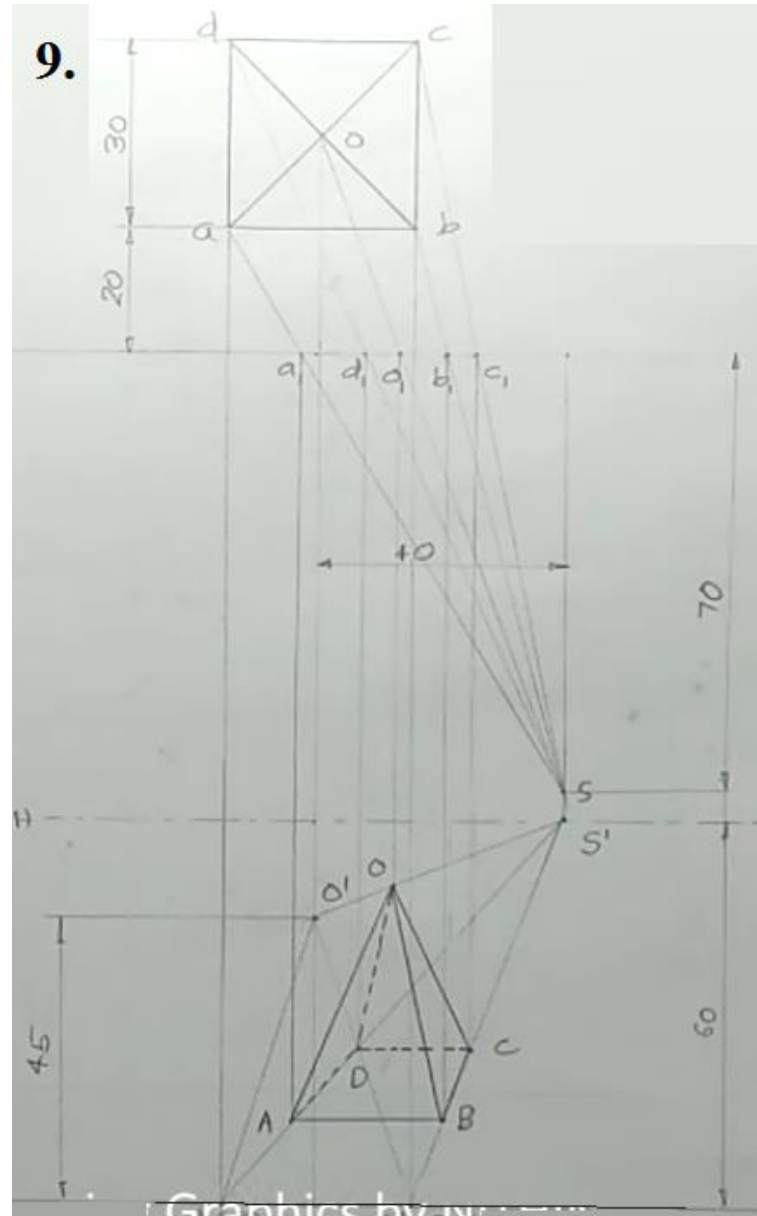
7.



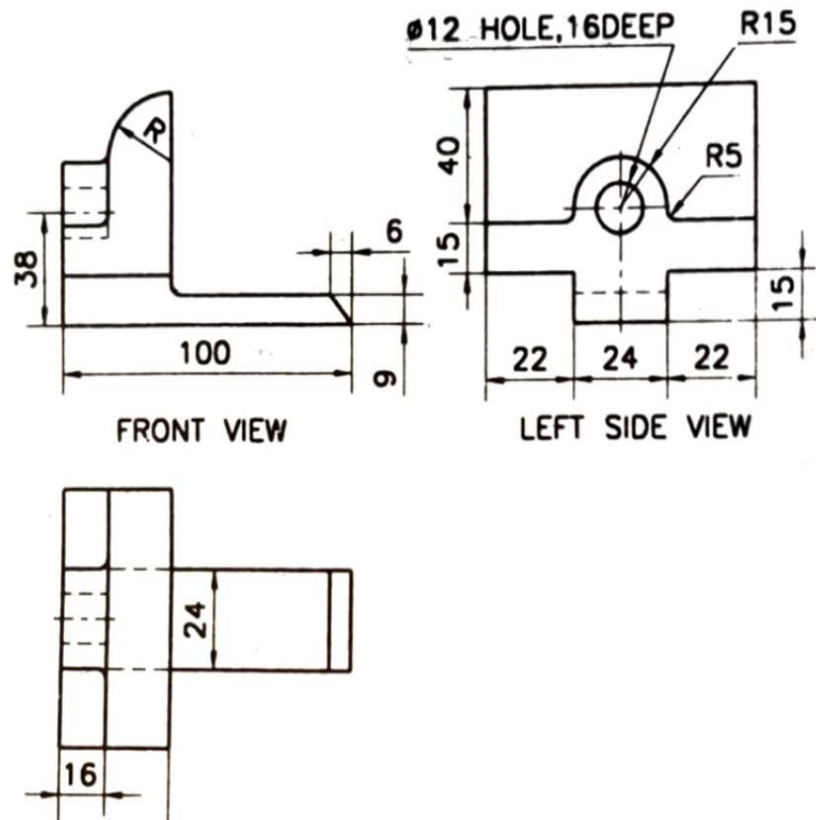
8.



9.



10.

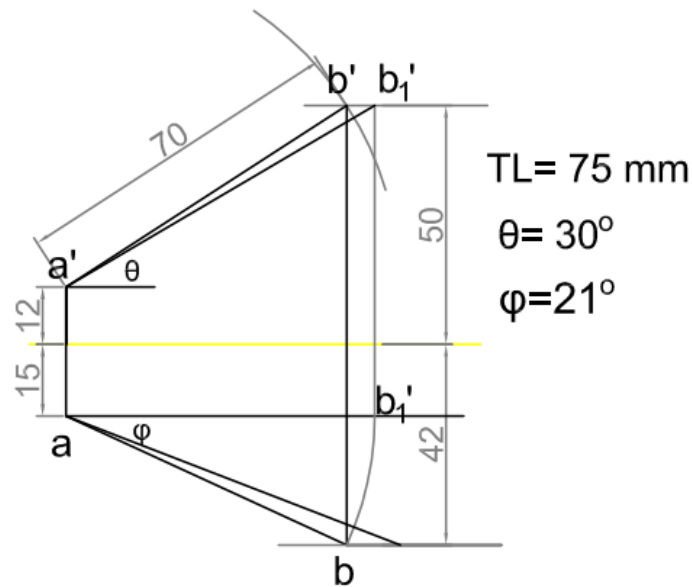


SCHEME FOR VALUATION

Solution provided here are indicative only. Candidates may not be penalised for small deviations from the values above. Correct answers arrived at using alternate methods should be accepted and marks awarded accordingly.

MODULE 1

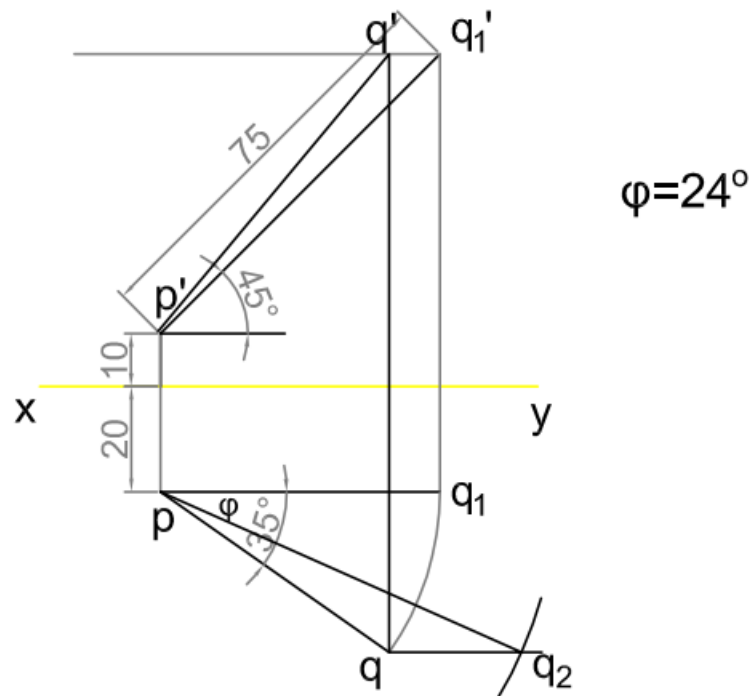
- One end point of a line AB is 12 mm above HP and is 15 mm in-front of VP. Other end point is 50 mm above HP and is 42 mm in front of VP. Draw the projections of the line AB if its elevation measures 70 mm. Find out its true length and the true inclinations with respect to the reference planes.



Scheme of Valuation:

No	Description	Marks
1	Drawing of XY line, locating the front view and top view of A	3
2	Locating the locus of the front view and top view of B	3
3	Drawing of the elevation of the line and locating the top view	3
4	Determining of the true length of the line	2
5	Determining the inclinations of the line with HP and VP	4
6	General Neatness	2
7	Proper Dimensioning	3
	TOTAL	20

- 2 One end point P of a line PQ, 75 mm long, is 10 mm above HP and 20 mm in front of VP. The line is inclined 45° to HP and its plan is inclined 35° to x-y line. Draw the projections of the line PQ and find out true inclination of the line with respect to VP.

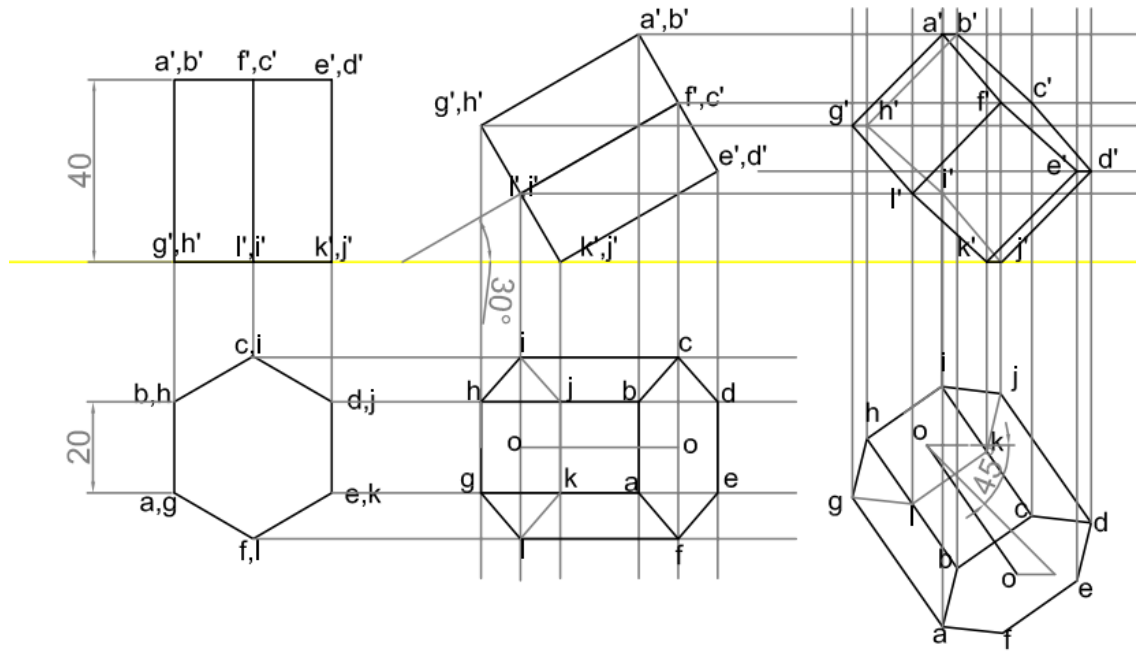


Scheme of Valuation:

No	Description	Marks
1	Drawing of XY line, locating the front view and top view of P	3
2	Drawing of the true length line at required angle	3
3	Drawing of the top view	3
4	Drawing the front view	2
5	Determining the inclinations of the line with VP	4
6	General Neatness	2
7	Proper Dimensioning	3
		20

MODULE 2

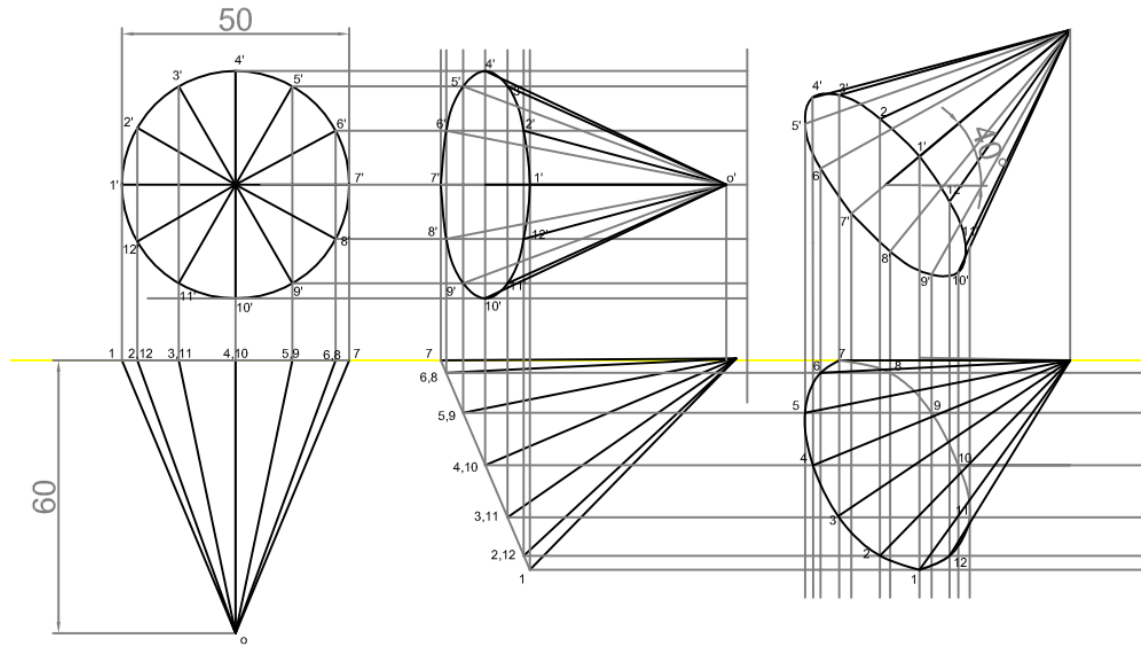
- 3 A hexagonal prism base 20 mm side and axis 40 mm long is placed with one of its base edges on the HP such that the axis is inclined at 30° to HP and 45° to VP. Draw the projections of the prism.



Scheme of Valuation:

No	Description	Marks
1	Drawing of the front and top views	4
2	Plan and Elevation after giving the axis inclination with HP	4
3	Obtaining the angle of top view of axis (either separately or in the same figure. The apparent angle of axis with VP obtained here is 55°)	2
4	Drawing the final top view	3
5	Drawing the final front view	3
6	General Neatness	2
7	Proper Dimensioning	2
		20

- 4 A cone of base diameter 50 mm and axis length 60 mm is resting on VP on one of its generators with the front view of the axis inclined at 40° to HP. Draw its projections.

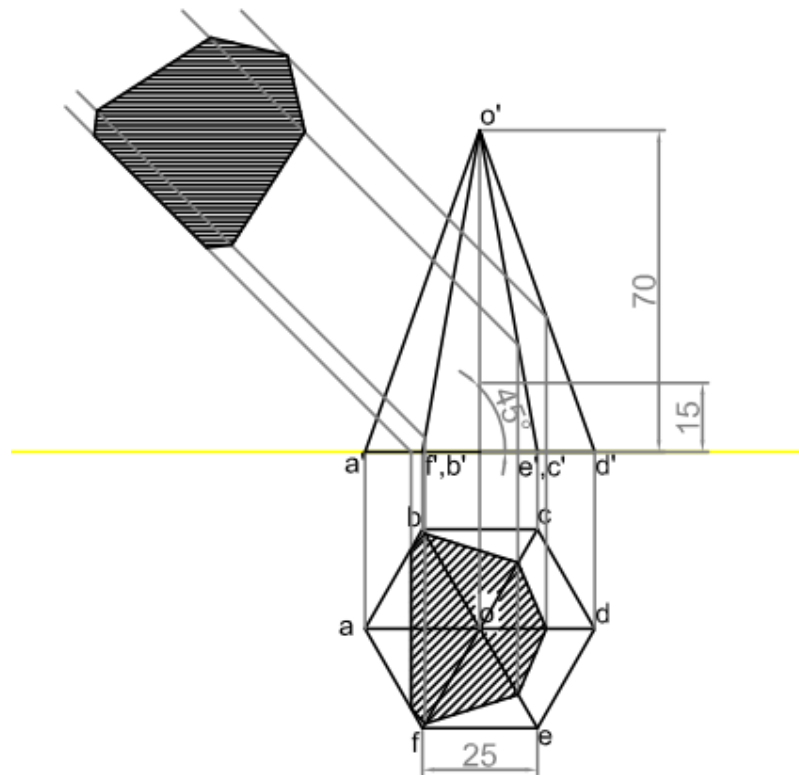


Scheme of Valuation:

No	Description	Marks
1	Drawing of the front and top views	5
2	Plan and Elevation after making the solid lie on its generator on VP	5
4	Drawing the final front view	3
5	Drawing the final top view	3
6	General Neatness	2
7	Proper Dimensioning and naming of various generators	2
		20

MODULE 3

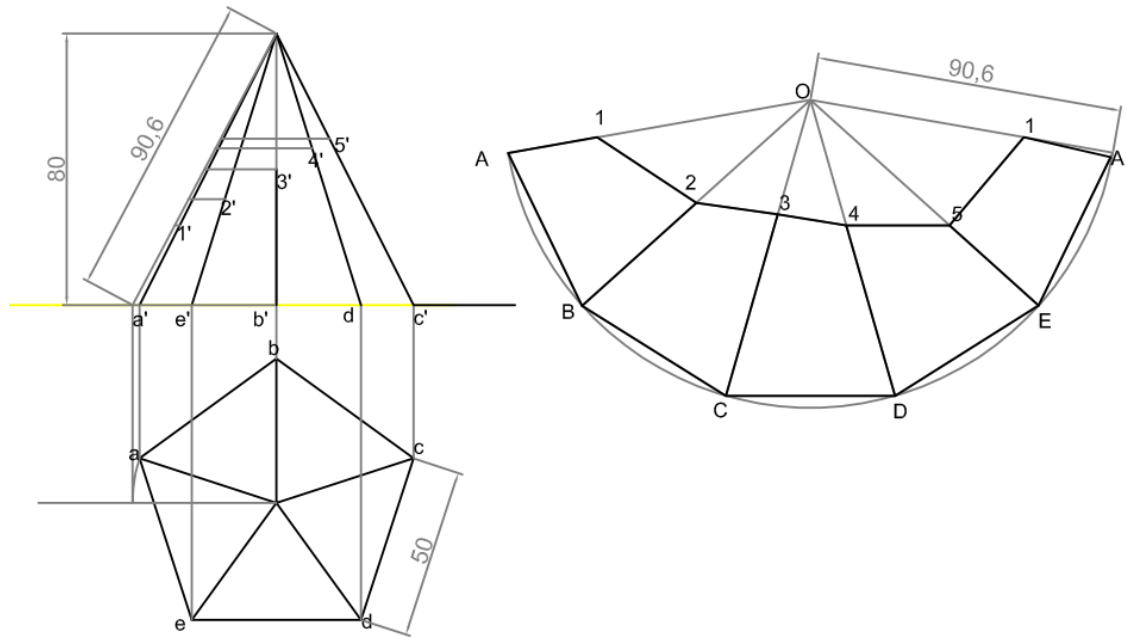
- 5 A hexagonal pyramid, side of base 25 mm and altitude 70 mm long, rests with its base on HP with two of its base sides parallel to VP. It is cut by a section plane perpendicular to VP, inclined at 45° to HP and passing through the axis 15mm from the base. Draw the sectional top view and true shape of the section.



Scheme of Valuation:

No	Description	Marks
1	Drawing of the front and top views	6
2	Marking of Section Plane	2
3	Sectional Top View	4
4	True Shape of the Section	4
5	General Neatness	2
6	Proper Dimensioning and naming of corners and edges	2
		20

- 6 A pentagonal pyramid, side of base 50 mm and height 80 mm rests on its base on the ground with one of its base sides parallel to VP. A section plane perpendicular to VP and inclined at 30° to HP cuts the pyramid, bisecting its axis. Draw the development of the truncated pyramid.

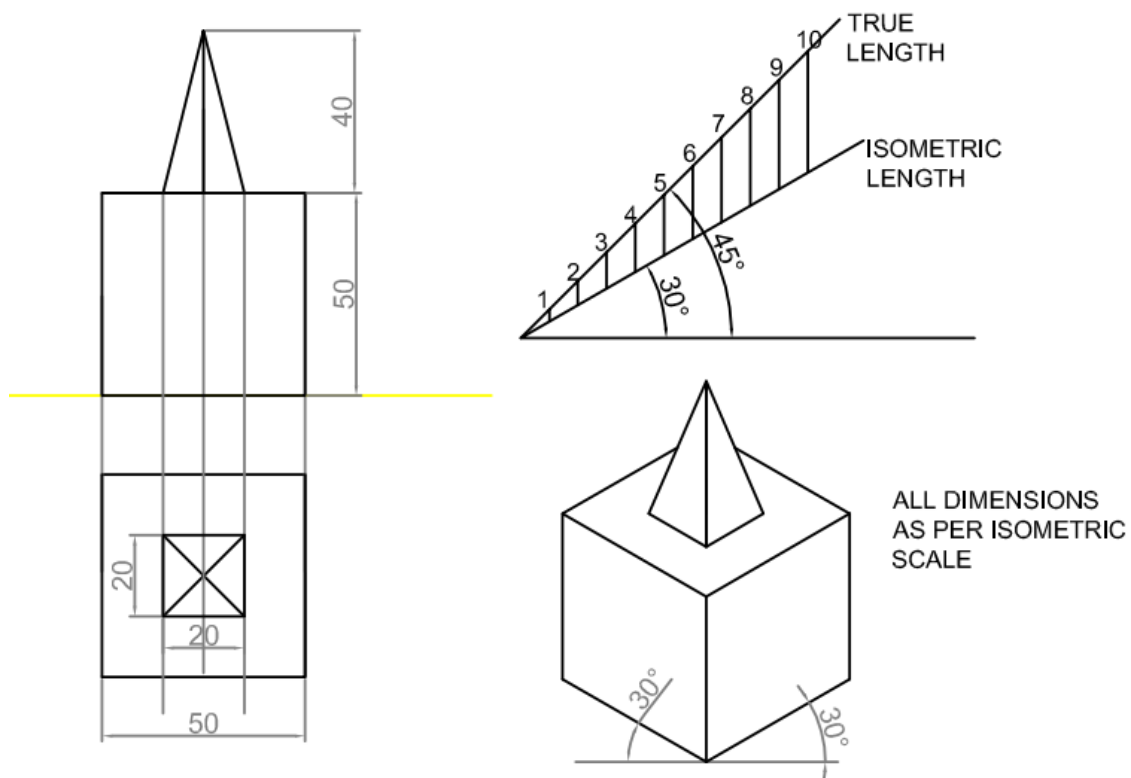


Scheme of Valuation:

No	Description	Marks
1	Drawing of the front and top views	6
2	Determination of actual length of the lateral edge	2
3	Drawing the development of lateral surface of the pyramid	4
4	Marking the corresponding section lines	4
5	General Neatness	2
6	Proper Dimensioning and naming of corners and edges	2
		20

MODULE 4

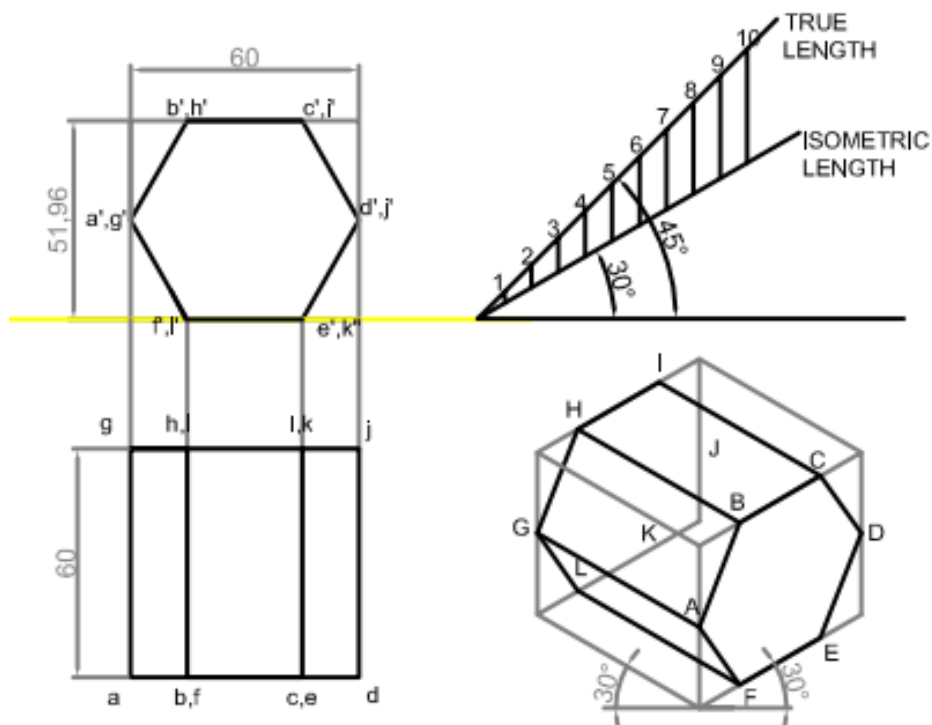
- 7 A square pyramid of base edge 20 mm and height 40 mm is mounted centrally on a face of a cube of base edge 50 mm. Draw the isometric projection of the objects.



Scheme of Valuation:

No	Description	Marks
1	Drawing of the front and top views	6
2	Construction of the Isometric Scale	4
3	Drawing the Isometric Projection of the Solids	8
4	General Neatness	2
		20

- 8 Draw the isometric projections of a hexagonal prism with edge of base 30 mm and axis 60 mm when it rests on a rectangular face on the ground.

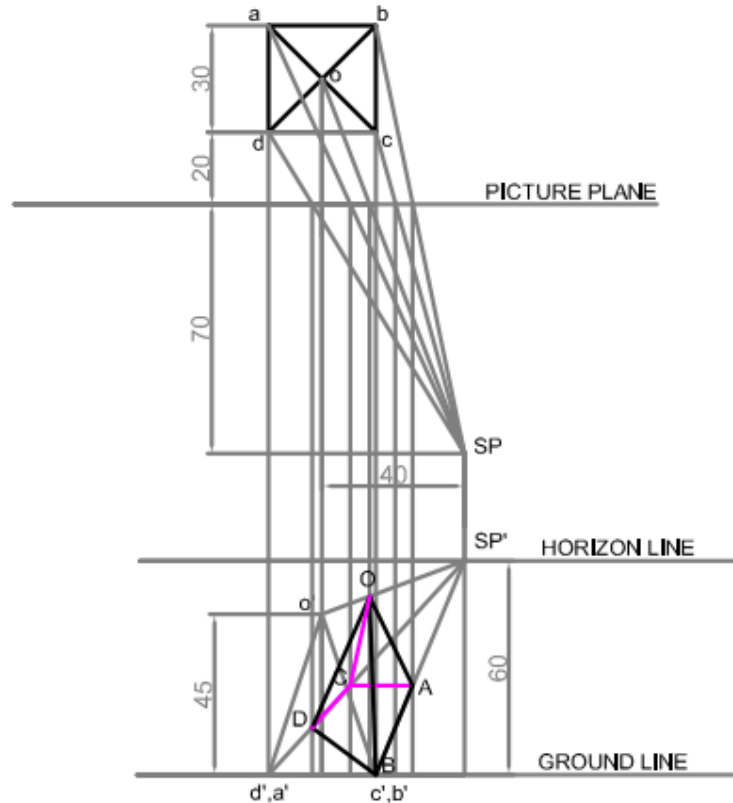


Scheme of Valuation:

No	Description	Marks
1	Drawing of the front and top views	6
2	Construction of the Isometric Scale	4
3	Drawing the Isometric Projection of the Solid	8
4	General Neatness	2
		20

MODULE 5

- 9 A square pyramid of base side 30 mm and height 45 mm is resting on the ground plane. The nearest edge of the base is parallel to and 20 mm behind the Picture Plane (PP). The station point is situated at a distance of 70 mm in front of the PP, 40 mm to the right of the axis of the pyramid, and 60 mm above the ground. Draw the perspective view of the pyramid.

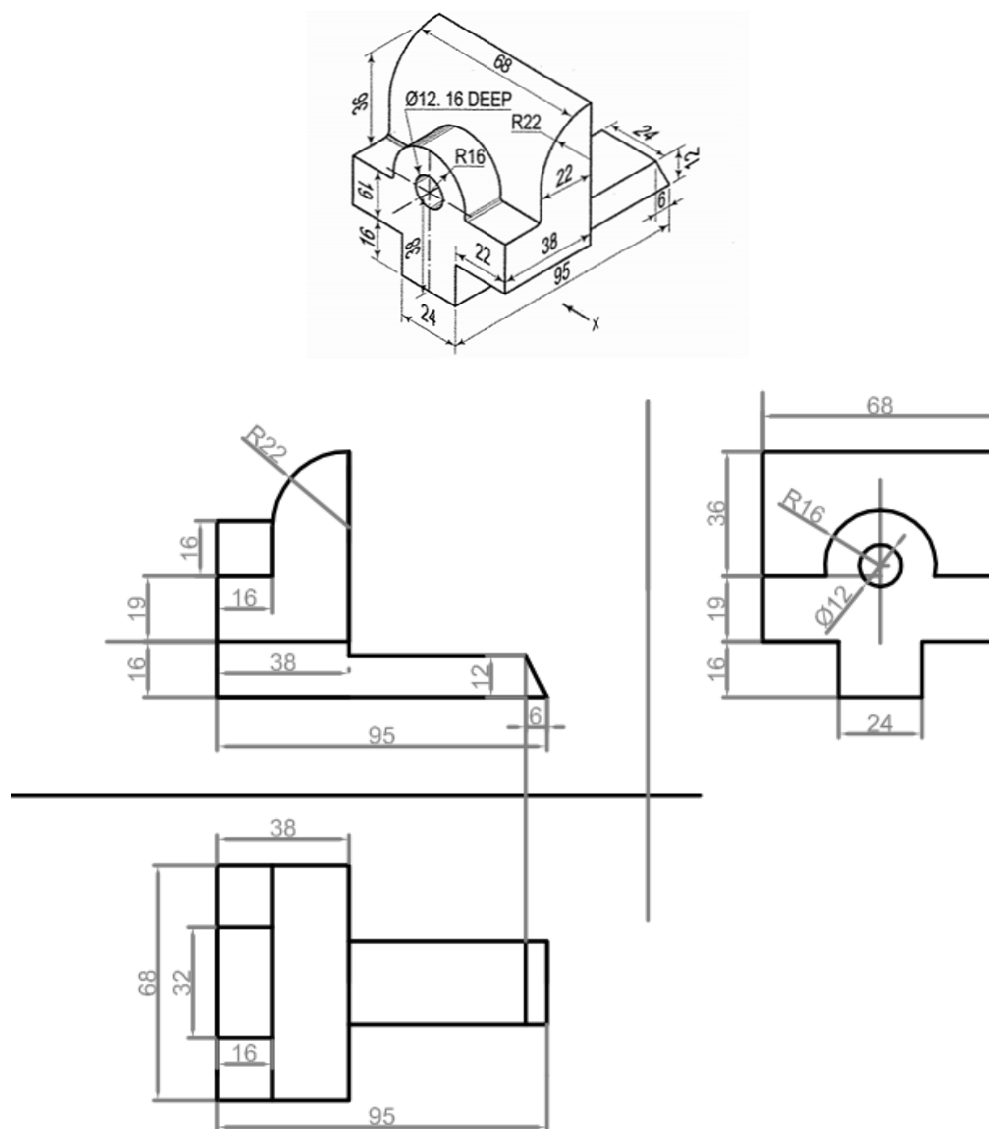


Note: Problem may also be solved with front view on top and top view below.

Scheme of Valuation:

No	Description	Marks
1	Drawing of front view and top views	6
2	Location of ground line, horizon line and the front and top views of the station points	4
3	Drawing of the perspective projection	8
4	General Neatness and dimensioning	2
		20

- 10 Draw the front view, top view, and side view of the object given below. Front view should be drawn as seen in the direction of the arrow X.



Scheme of Valuation:

No	Description	Marks
1	Drawing of the front view	6
2	Drawing of the top view	6
3	Drawing of the end view (Any one side view. Left side view is given above)	6
4	General Neatness and positioning of figures (It is expected that the three views are drawn as per first angle projection)	2
		20